MAHATMA GANDHI UNIVERSITY, KOTTAYAM

MGU-UGP (HONOURS)
THIRD SEMESTER EXAMINATION
(2024 ADMISSION ONWARDS)

Course Code: **MG3DSCBOT200** Course Title: **MICROBIOLOGY AND PHYCOLOGY**

Time: 1.5 Hrs **Maximum Marks: 50**

Remember (K), Understand (U), Apply (A), Analyse (An), Evaluate (E), Create (C), Skill (S), Interest (I) and Appreciation (Ap)

Student should attempt at least one question from each course outcome to enhance their overall outcome attainability

Part **A**: Short answers
Answer 10 questions
Each question carries 1 mark

	Questions	Taxonomy	CO
1	What are plasmids?	Remember	1
2	Give an example for a disease caused by prions.	Understand	1
3	Name a microbe used as biocontrol agent.	Remember	1
4	Identify a microorganism commonly utilized as a food source.	Remember	1
5	Who proposed the modern system of algal classification?	Remember	2
6	Distinguish isogamy from oogamy.	Understand	2
7	Outline any two characteristic features of coenobium in *Volvox.*	Analyze	2
8	What type of chloroplast is found in *Spirogyra*?	Remember	2
9	Compare haplodiplontic and diplontic life cycle?	Evaluate	2
10	Name any two commercial products from red algae.	Remember	3
11	Expand SCP.	Understand	3
12	What are algal blooms?	Remember	3

Part **B**:
Short Essay Type Questions
Answer 6 Questions
Each question carries 5 marks

13	Explain the structure of a bacteriophage.	Understand	1
14	Write a short note on the industrial and agricultural applications of microbes.	Understand	1
15	List the different modes of asexual reproduction in algae.	Remember	2
16	Outline the pigment composition and photosynthetic end products in major algal groups.	Analyze	2
17	Compare and contrast the thallus structures of *Volvox* and *Spirogyra.*	Analyze	2
18	Describe the characteristic features of *Sargassum.*	Remember	2
19	Algal associations are ecologically significant. Justify	Understand	3
20	Summarize the methods of algal cultivation.	Understand	3

Part **C**:
Essay Type Questions
Answer any one question
Question carries 10 marks

21	Explain the different mechanisms by which genetic recombination occurs in bacteria.	Understand	1
22	Write an essay on the beneficial aspects of algae.	Understand	3